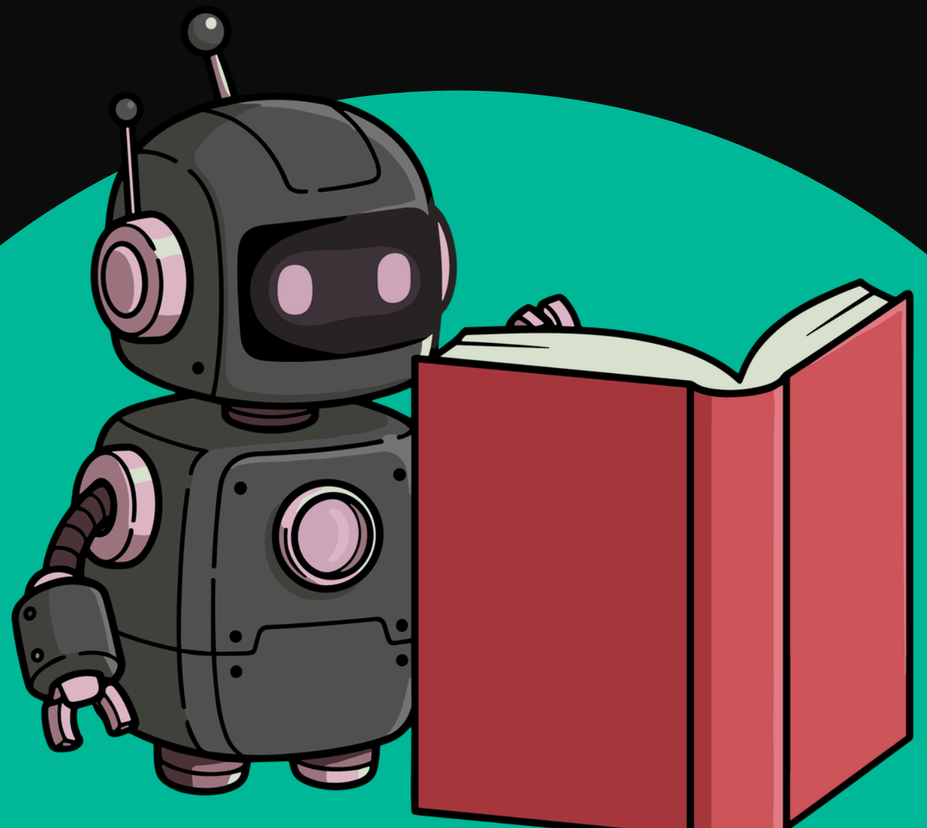
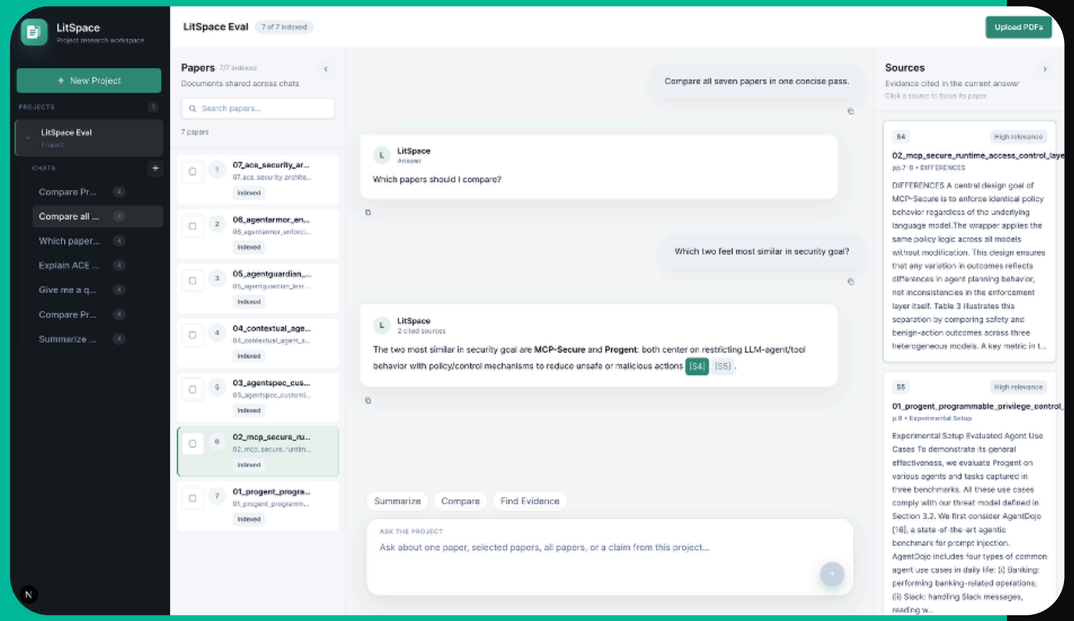
LitSpace

A grounded multi-paper research assistant for academic PDFs

Presented by Enas Batarfi



The Main Challenge is Fluent ≠ Grounded



project-bounded · source visible · multi-paper

- Answers may rely on outside model knowledge
- Claims can get mixed across papers
- Users still ask, “Where exactly is the evidence?”
- Research needs source-bounded, verifiable answers

Our Approach

1. Upload PDFs
2. Parse + chunk
3. Hybrid retrieval
4. Route action
5. Grounded answer
6. Show sources

Key Design Decisions

Project Scope	Answers only from uploaded papers
Hybrid Retrieval	Embeddings + BM25 + rank fusion
Structured Chunking	Page, section, and overlap metadata
Answer Routing	Answer, clarify, or refuse

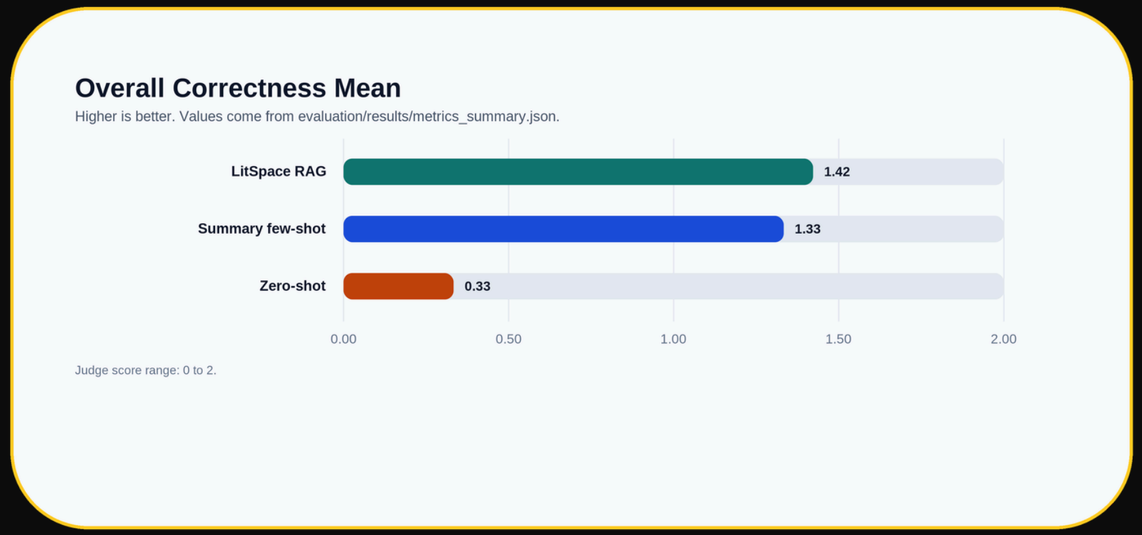
Evaluation & Limitations

45 benchmark questions, 3 systems compared

Zero-shot
no retrieval

Summary few-shot
paper summary cards

LitSpace RAG
retrieval + routing + sources



1.42
LitSpace correctness

75.6%
win rate vs zero-shot

48.9%
win rate vs summary

0.97
paper hit@5

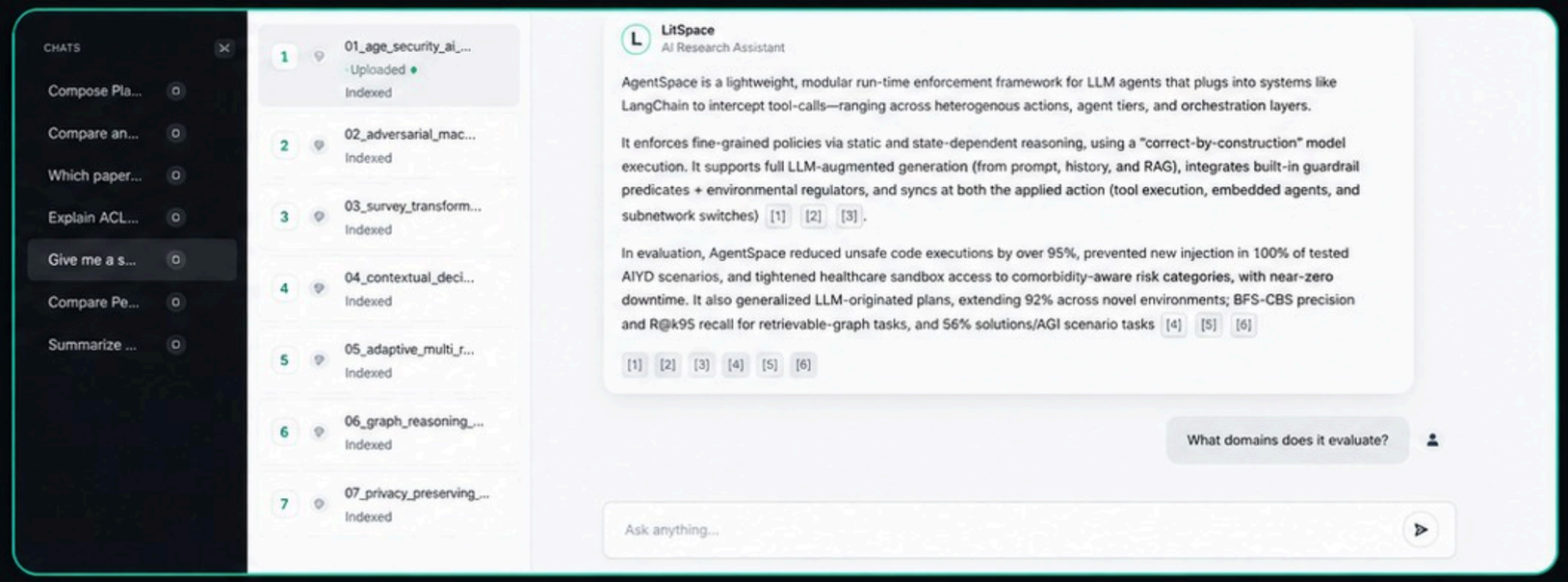
0.31
section recall@5

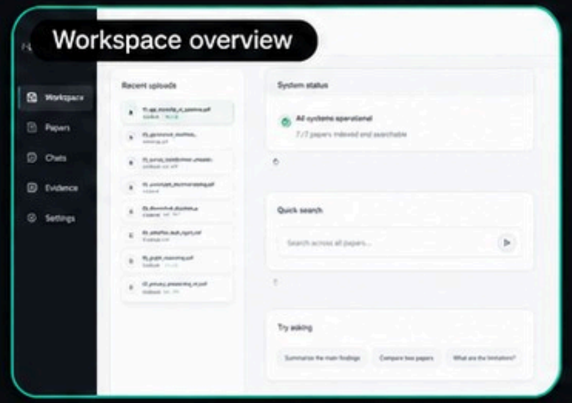
Live Demo: LitSpace in Action

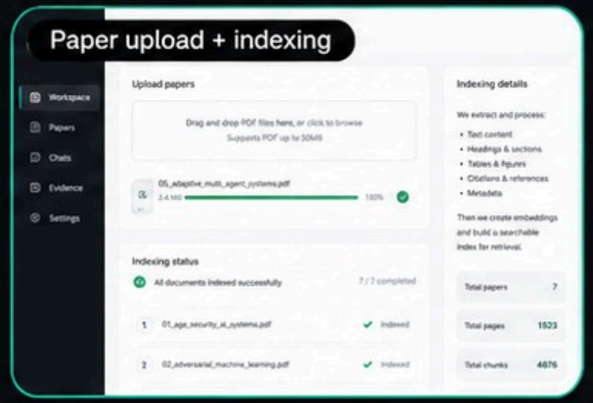
1 Upload + index
Seven research papers become searchable.

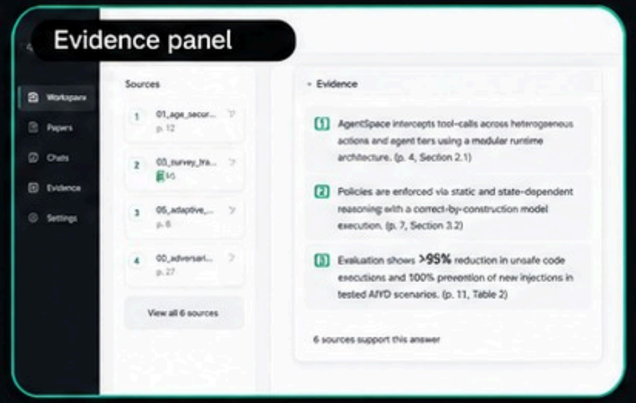
2 Ask a question
The system retrieves context before answering.

3 Verify evidence
Citations and source text stay visible.









Thank you for your attention!

Open to feedback and discussion.

https://github.com/EnasBatarfi/litspace